Code No: 132AA **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech I Year II Semester Examinations, March/April - 2023
ENGINEERING PHYSICS – II
(Common to EEE, ECE, CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART- A

(25 Marks)

1.a)	State and explain Heisenberg uncertainty principle.	[2]
b)	What is the importance of Davisson and Germer experiment?	[3]
c)	What is the Fermi level?	[2]
d)	Explain the working principle of a Solar cell.	[3]
e)	Define electric dipole, polarizability.	[2]
f)	Define piezoelectricity and pyroelectricity.	[3]
g)	Explain the Permeability in magnetism.	[2]
h)	Mention three applications of superconductivity.	[3]
i)	Mention two biomedical application of nanotechnology.	[2]
j)	Explain quantum confinement phenomenon?	[3]

PART-B

(50 Marks)

2.a) What are the matter waves? Derive an expression for the de-Broglie wavelength of an electron.
b) Explain energy band formation in solids. [5+5]

OR

3.a) Explain the behavior of an electron moving in a field of periodic potential using kronig and Penny model.
b) Explain in detail E-K curve. [5+5]

4.a) Calculate the carrier concentration in P-type semiconductor.
b) Explain direct and indirect band gap semiconductors. [5+5]

OR

5.a) Explain the formation of PN junction diode and V-I characteristics of the same.
b) Derive an expression for density of electrons in intrinsic semiconductors. [5+5]

6.a) Derive an expression for ionic polarizability.
b) Derive Clausius - Mossotti relation. [5+5]

OR

7.a) What is meant by local field in dielectrics and how it is calculated for cubic structure?
b) What is meant by Ferro electricity? List out the applications of ferro-electrics. [5+5]

8.a) Explain in detail 'Meissner effect' and magnetic levitation.
b) Write notes on Type-I, and Type-II super conductors. [5+5]

OR

9.a) Discuss the properties and effect of external field on the para and ferro magnetic materials.
b) What is meant by Hysteresis loss? Describe the formation of hysteresis loop using domain wall movement. [5+5]

10.a) Explain in detail Sol–gel preparation method. Discuss the applications of Nanotechnology.
b) How do you determine the size of the particle using XRD characterization technique? [5+5]

OR

11.a) How do you synthesize the nonmaterial using Chemical Vapor Deposition (CVD) method?
b) Write short note on Ball milling method. [5+5]

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